# 2-ACRE MORINGA COMMERCIAL ORCHARD — IRRIGATION ENGINEERING MASTER BLUEPRINT

HYDRAULIC DISTRIBUTION NETWORK • OVERHEAD DOUBLE-TEE RISERS • INLINE & ONLINE DRIP LAYOUT

STANDARDS: IS 12786 (DRIP LATERALS) | IS 4985 (PVC PRESSURE PIPELINES) | DESIGN PRESSURE: 2.5 KG/CM²

CENTRAL ANNOTATION CHANNEL (ZERO-OVERLAP)
165'-0" (PLOT B WIDTH)
PLOT B: BHAGYA MORINGA (1 ACRE | 7ft x 8ft GRID | 750 TREES)
32 ROWS (WESTWARD ONLY) • 16MM PLAIN LATERALS • 24 PUNCHED TUNNING DRIPPERS/ROW (ITEM #2)
40MM FLUSH VALVE (NORTH TERMINAL)
ELEVATION: -2'-0" IN SURFACE SLEEVE
COVERS TOP 4 ROWS (ROWS #29-32)
RISER B4 (OFFSET: 248'-0" NORTH)
2" MAINLINE TERMINATES HERE (2" END CAP)
OVERHEAD 2" TEE + 2x 40MM BALL VALVES
ONLY 1.25" SUBLINE CONTINUES 32 FT NORTH
RISER B3 (OFFSET: 184'-0" NORTH)
4FT DOUBLE-TEE + 2x 40MM BALL VALVES
1.25" SUBLINES: 32' SOUTH & 32' NORTH
FEEDS 8 ROWS (ROWS #17-24 | 192 TREES)
256'-0" ACTIVE HEIGHT
32 ROWS @ 8'-0" SPACING
RISER B2 (OFFSET: 120'-0" NORTH)
4FT DOUBLE-TEE + 2x 40MM BALL VALVES
1.25" SUBLINES: 32' SOUTH & 32' NORTH
FEEDS 8 ROWS (ROWS #9-16 | 192 TREES)
145'-0" CENTER MAINLINE
24 TRENCHES @ 5' SPACING
120 FT ACTIVE SUBLINE
RISER B1 (OFFSET: 56'-0" NORTH)
4FT DOUBLE-TEE + 2x 40MM BALL VALVES
SOUTH ARM EXTENDS 32' TO PLOT B START (24')
FEEDS 8 ROWS (ROWS #1-8 | 192 TREES)
PLOT B START BOUNDARY (24'-0" FROM TEE B)
40MM FLUSH VALVE (SOUTH ARM TERMINAL)
ROW #1 STARTS HERE (NO ROWS BELOW THIS LINE)
24' BARE GROUND RUN
(NO PLANTS / NO ROWS)
2" END CAP
TEE B: PLOT B ENTRY
COORD: (285', 100')
2" TEE WITH WEST CAP
390'-0" (PLOT A OVERALL TRENCH WIDTH: 195' W + 195' E)
PLOT A: PKM-1 ANNUAL MORINGA (1 ACRE | 3ft x 5ft HIGH-DENSITY | 3,000 PLANTS)
24 ACTIVE TRENCHES @ 5' SPACING • 16x4x40 INLINE DRIP LATERALS • 50x16x2-WAY TAKE-OFF COLLARS (V-CONNECTORS)
16x4x40 INLINE LA
390 FT TOTAL TREN
195 FT WEST + 19
16MM ENDCAPS ON
FEEDS 6 TRENCHES
RISER A3 (OFFSET
OVERHEAD 2" TEE
1.5" SUBLINES: 1
FEEDS 6 TRENCHES
RISER A2 (OFFSET
OVERHEAD 2" TEE
1.5" SUBLINES: 1
FEEDS 6 TRENCHES
RISER A1 (OFFSET
OVERHEAD 2" TEE
1.5" SUBLINES: 1
FEEDS 6 TRENCHES
215'-0" 2" MAINLINE RUN (UNDER PLOT A & ROADWAY BUFFER)
TRUE PIPE LENGTH: 215 FT | TEE A (500', 100') TO TEE B (285', 100')
2" END CAP
90'-0" 2" PVC MAINLINE (63MM)
TEE A: PLOT A ENTRY

PROJECT & LOCATION:

**2-ACRE MORINGA COMMERCIAL ORCHARD DRIP SYSTEM**

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| DRAWING TITLE:<br>**MASTER FIELD BLUEPRINT** | ZONE BREAKDOWN:<br>**ZONE 1: 5HP SOLAR BOREWELL (2" MAINLINE)**<br>ZONE 2: PLOT A (PKM-1, 24 TRENCHES, 3000P) | ZONE 3 (PLOT B SPEC):<br>**BHAGYA (7X8 FT, 32 ROWS, 750P)**<br>MAINLINE TERMINATES AT RISER B4 |
| HYDRAULIC CRITERIA:<br>**DESIGN VELOCITY: < 1.5 M/S**<br>HEAD LOSS: < 2.0 M / 100M | FITTINGS SCHEDULE:<br>**16x PLAIN BALL VALVES**<br>16x FLUSH VALVES (SURFACE) | EMITTER SPECIFICATION:<br>**PLOT A: 4 LPH INLINE @ 40CM**<br>PLOT B: TUNNING DRIPPER @ 7FT |

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| SCALE:<br>**1 FT = 5 PX**<br>(1:60 ON A0 / ARCH E) | DATE / STATUS:<br>**OCTOBER 2026**<br>APPROVED FOR FIELD | CAD STANDARD:<br>**IS 12786 / 4985**<br>CLASSIC BLUEPRINT | SHEET NUMBER:<br>**SHEET 1 OF 2** |